

CSA1304-THEORYOF COMPUTATION LAB EXPERIMENTS

5. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0S0 \mid A$ $A \rightarrow 1A \mid \epsilon$

AIM

Write a C program to check whether a given string belongs to the Context-Free Grammar (CFG).

CODE

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[100];
    int i, j, len;

    printf("Enter the binary string: ");
    scanf("%s", str);

    len = strlen(str);

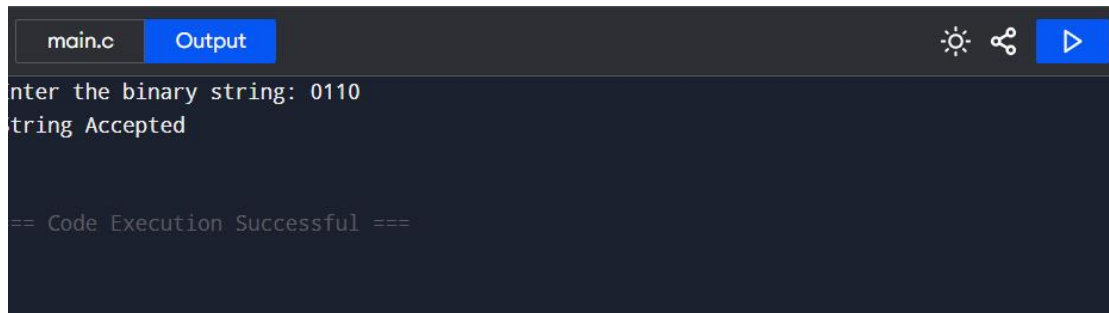
    i = 0;
    j = len - 1;

    while(i <= j && str[i] == '0' && str[j] == '0')
    {
        i++;
        j--;
    }

    while(i <= j)
    {
        if(str[i] != '1')
        {
            printf("String Rejected\n");
            return 0;
        }
        i++;
    }
```

```
}  
  
printf("String Accepted\n");  
  
return 0;  
}
```

OUTPUT



```
main.c  Output  
Enter the binary string: 0110  
String Accepted  
  
== Code Execution Successful ==
```